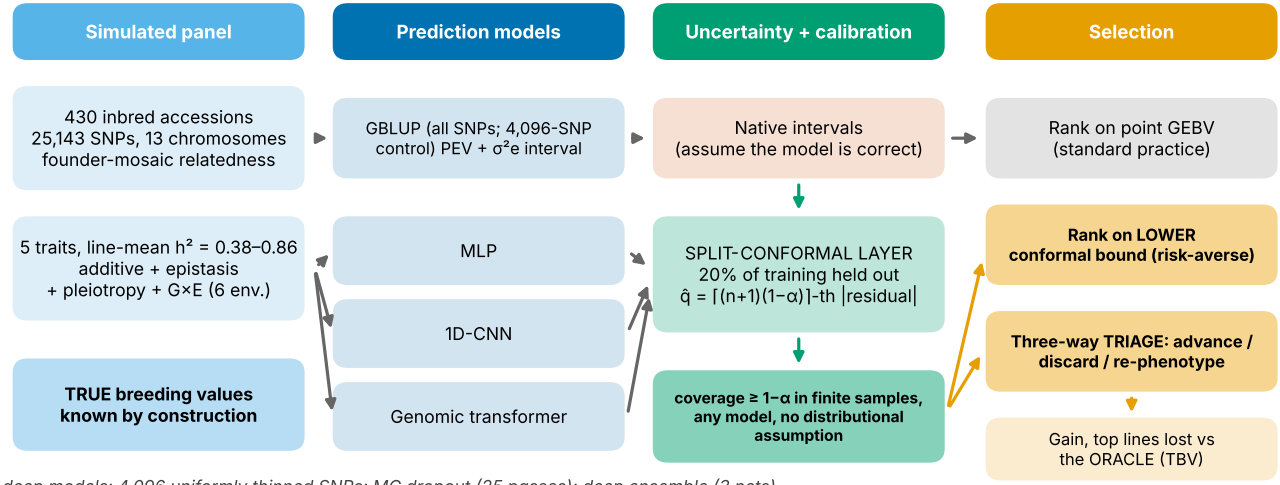


deep models: 4,096 uniformly thinned SNPs; MC dropout (25 passes); deep ensemble (3 nets)

robustness: 5 independent panels; TBV coverage; unseen subpopulation (LOSO) and environment (LOEO); 80 to 800 training lines

The four questions

i. How accurate are the models, at 80 to 800 training lines, relative to the heritability ceiling?

ii. Are their native uncertainty statements calibrated, for the phenotype and for the breeding value?

iii. Does split-conformal calibration restore coverage, at what width, and under subpopulation or environment shift?

iv. Can calibrated intervals improve a selection decision?